Do Now · retrieval

Year 13 autumn mock: Paper 1

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Evaluate true AND false.

Q2

Why should a model omit irrelevant details?

Q3

What does the Program Counter hold?

Q4

A full adder receives A=1, B=1 and carry-in=1. State sum and carry-out and justify using binary addition.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Purpose — This is an original diagnostic Paper 1 practice, not an official OCR examination paper. Some systems topics will be retaught later.

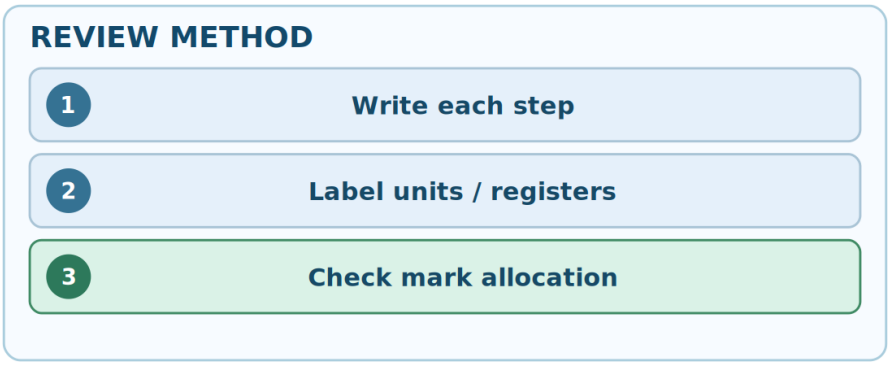
Core idea 2

Manage time — Allow approximately 15 minutes to each 15-mark question.

Core idea 3

Show methods — Write the inversion/add-one steps, state every Boolean simplification rule and use precise CPU register terminology.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

diagnostic — Assessment used to find knowledge gaps.

two's complement — Encoding signed binary integers.

De Morgan's law — Negate operands and swap AND/OR.

register — Fast CPU storage for a specific role.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

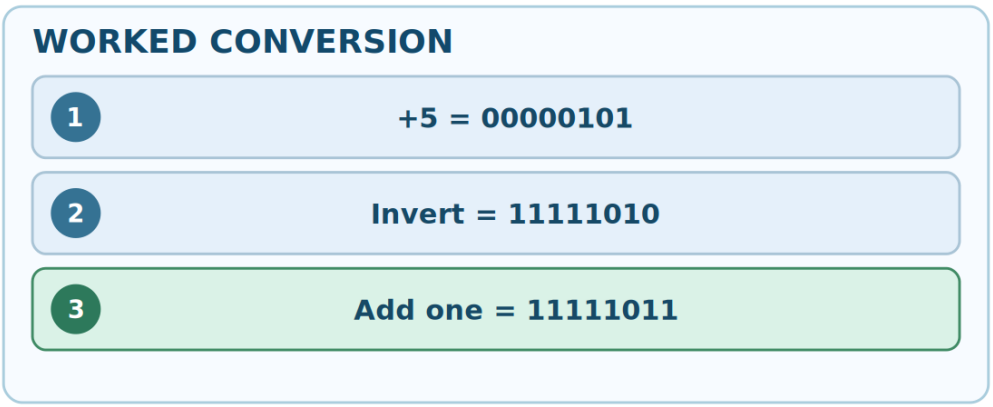
Problem

Unmarked example: what earns method marks for a negative 8-bit value?

Follow the worked method

1. Predict the method — Write +5 in exactly 8 bits.
2. Check and explain — Invert each bit, then add one to obtain the two's complement result for −5.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict the method
First calculate A AND B = 0.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

(a) Convert −18 to eight-bit two's complement, showing working. [4] (b) Perform arithmetic right shift by 1 on that representation, giving the binary result and signed decimal value. [3] (c) Add 1011_2 and 0110_2, showing carry and answer. [3] (d) Normalise positive $0.0011_2 \times 2^5$, showing shifts and decimal value. [5]

Question 2

(a) Complete four output rows of A XOR B in order 00,01,10,11. [4] (b) Simplify ¬(A ∧ B) using De Morgan's law. [3] (c) For a half adder with A=1, B=1, state Sum and Carry and explain the gates. [4] (d) Simplify A ∨ (B ∧ ¬B) using Boolean identities. [4]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

(a) State the roles of PC, MAR, MDR, CIR and ACC. [5] (b) Describe a simplified fetch-decode-execute cycle using PC, MAR, MDR, CIR and the control unit. [6] (c) Explain one role of an interrupt and an interrupt service routine. [4]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

For an 8-bit signed representation, adding a ninth bit automatically gives the correct 8-bit answer.

Use key vocabulary from memory

1. Explain diagnostic in your own words.

2. Explain two's complement in your own words.

3. Explain De Morgan's law in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.